

ONE ID. CONFLICTING SOURCES

W4385245566

A real public work record combines the DOI and authors of one paper, the title of another, and locations for a third publication family.

OBSERVED OPENALEX TITLE

Exploiting Generative AI to Scale up Intelligent Tutoring Systems

TITLE REGISTERED FOR ITS PRIMARY DOI

MizAR 60 for Mizar 50

10.4230/lipics.itp.2023.19

5	3	9/9
DOI LOCATIONS	TITLE FAMILIES	AUTHOR ORCIDS MATCH

What is built: a free, read-only source audit that exposes conflicts and exports the evidence needed to review them. Detection and evidence assembly are implemented; no upstream correction is claimed.

THINK OF A LIBRARY CATALOG CARD

A catalog card tells you a book's title, who wrote it, and where to find it. OpenAlex does something similar for research papers. A DOI acts like a persistent barcode for a particular research object.

Now imagine one card carrying **Book A's barcode and authors, Book B's title, and links to Book C**. Correcting the printed title would make the card look better, but the mixed-up links would still be there.

WHAT WE ACTUALLY FOUND

The real record W4385245566 has the DOI and nine authors of **MizAR 60 for Mizar 50**. Its title is **Exploiting Generative AI to Scale up Intelligent Tutoring Systems**, a different paper with two different authors. It also links to a hardware-compression paper by another author.

There are five DOI links: two Dagstuhl papers and three linked records for the hardware paper. The hardware records explicitly relate its versions. That relationship is legitimate. The concern is the three different publication families appearing on one work record.

WHAT SOURCECHECK DOES

You give it a work ID or DOI. It checks the attached DOI records against public registration metadata, shows which source says what, and saves the evidence. For this case, it gives an engineer a clear reason to review the source associations instead of guessing a replacement title.

WHY IT STOPS BEFORE EDITING

A safe correction may require separating works and reviewing their citations or other derived data. This tool has neither the internal history nor authority to make those decisions. It supplies a review packet to the people who do. It never writes to OpenAlex.

The practical benefit to test: less repeated source lookup and a handoff another engineer can reproduce. The detection works on the captured case. Time saved and usefulness in a real support queue have not yet been measured.

INSPECT THE REAL RECORD

THREE FAMILIES, ONE RECORD

Every DOI below is present in the captured OpenAlex locations. Titles and creators come from each DOI's DataCite record. The two Dagstuhl publisher pages corroborate their own DOI, title, and creator metadata.

A / 2023

MizAR 60 for Mizar 50

10.4230/lipics.itp.2023.19

Jan Jakubův + 8 coauthors

Primary DOI. All nine OpenAlex author names and all nine ORCIDs match this family.

B / 2026

Exploiting Generative AI to Scale up Intelligent Tutoring Systems

10.4230/oasics.icpec.2026.2

Miguel Ferreira; José Paulo Leal

This title exactly matches OpenAlex's observed title. No creator ORCID overlaps family A.

C / 2026

Inline Hardware KV-Cache Compression for Long-Context Transformer Inference: An Architectural Case for a Memory-Path Compression Engine

10.5281/zenodo.21230429 / .21230430 / .21230434

Alexander Novickis

Three DOI records with explicit HasVersion / IsVersionOf relations. No creator ORCID overlaps A or B.

MULTIPLE DOIS ARE NOT AUTOMATICALLY AN ERROR

The Zenodo links are a useful control: explicit version relations explain why its three DOI records belong to a family. They do not link that family to either Dagstuhl paper. Titles and author sets reinforce the distinction, but the family split remains a review proposal.

WHY A TITLE-ONLY PATCH IS INSUFFICIENT

Overwriting the title would leave unrelated locations attached. A maintainer must examine the source associations and any affected citations, topics, or extracted text. This investigation has not established which downstream fields need rebuilding.

MAKE THE CONFLICT REVIEWABLE

SourceCheck accepts a DOI or OpenAlex work ID. The Python CLI runs locally; the shared site performs live checks in the browser. Both preserve a clear boundary between source disagreement and a decision to edit a scholarly record.

01 CAPTURE

Fetch the OpenAlex work and at most six selected, canonical, or location DOI records. Query Crossref first; fall back to DataCite after a Crossref 404.

02 COMPARE

Compare registered titles and creator context per DOI. Keep typed relationships visible. A citation, reference, or container link is not an identity claim.

03 CLASSIFY

Return aligned, review, or inconclusive. An aligned result covers the checked evidence only. Missing sources, failed requests, and truncated coverage are limitations.

04 EXPORT

Produce a read-only review packet with source claims and HTTP evidence. It gives a maintainer a reproducible starting point without guessing a global replacement title.

A LEAD TO INVESTIGATE, NOT A PROVEN ROOT CAUSE

All five DOI records contain a **Cites** relation to **arXiv:1706.03762**. One hypothesis is that a cited identifier was treated as an identity signal during ingestion or merging. A shared citation alone does not prove that happened.

The invariant is still clear: a Cites edge must not, on its own, equate works. DataCite distinguishes citation relationships from identity and version relationships. Internal production logs are needed to establish the cause.

DATAHITE RELATIONSHIP SEMANTICS

FROM REPORT TO REVIEWED CORRECTION

Illustrative ticket using the real record: “The DOI on W4385245566 points to the MizAR paper, but the title is about tutoring.” This is a walkthrough, not a ticket that was submitted or handled by OpenAlex.

- 01 PASTE THE ID**

The support engineer enters W4385245566. The captured case is available immediately; the engineer chooses Check live to inspect the current public record.
- 02 CHECK THE SOURCES**

The tool fetches the work and up to six selected DOI records. It records responses and reports review, aligned, or inconclusive. Fetch failures and incomplete coverage remain visible.
- 03 READ THE CONFLICT**

The engineer compares three title families and checks the legitimate Zenodo version links. Creator names and ORCIDs add context; the automated outcome is based on registered main-title comparisons, not an author-identity decision.
- 04 EXPORT THE HANDOFF**

The engineer downloads the JSON report with its receipts and the Markdown triage note. Another person can inspect the claims, follow the source URLs, and verify the captured response hashes.
- 05 MAINTAINER REVIEW / FUTURE HUMAN ACTION**

An authorized maintainer checks internal source history, decides whether a split or other correction is justified, and uses OpenAlex's existing process. SourceCheck does not perform this step.
- 06 RECHECK / AFTER A FUTURE CORRECTION**

After any approved change is deployed, the engineer runs the same ID again and compares a new receipt with the original. A clean title check alone does not certify that all affected data has been repaired.

Steps 1-4 use the implemented artifact. Steps 5-6 describe a possible follow-through; no upstream correction or end-to-end support outcome has been demonstrated.

WOULD THIS HELP THE TEAM?

OpenAlex already documents agents that read tickets, verify proposed corrections against evidence, apply them, or escalate. SourceCheck is a small external companion: deterministic comparisons and a portable evidence packet. It is not a new curation platform or evidence that the team lacks these capabilities.

OPENALEX'S EXISTING CORRECTION WORKFLOW

WHERE IT CAN FAIL

False flags: legitimate versions, translations, alternate titles, or registry updates can make titles differ. A reviewer must inspect the relationships. A differing title is not proof that sources belong to different works.

Missed problems: a same-title record with wrong authors can pass this title-led audit. Creator context is not a full identity model. Records without DOIs, non-DOI locations, and sources beyond the six-DOI cap are outside a complete check.

Illustrative rule limits: stripping punctuation makes “C++ for Biology” and “C for Biology” compare equal. A separately stored subtitle can create a harmless title difference. These are analytical counterexamples, not observed OpenAlex defects. Publisher-page and nine-ORCID verification are case-specific evidence, not general live-check features.

Evidence limits: registry metadata can be wrong and may come from the publisher's own feed. Agreement is not independent proof. The internal cause of this case is unknown. No support-time saving, detector precision, or production reliability has been measured.

PROPOSED PILOT / NOT RUN

Take **20 consecutive eligible DOI-bearing metadata tickets**. Using predefined complexity criteria, form ten matched pairs. Randomly assign one ticket per pair to the team's current agents and tooling, and one to that same process plus SourceCheck: ten per method. Keep every included case and failure in the results.

Balance engineer assignments across methods; no engineer handles the same ticket twice. Have independent adjudicators review standardized, de-branded outputs without knowing the method. This avoids a same-ticket crossover where the first investigation teaches the second.

Measure **active engineer minutes**, **useful newly confirmed findings**, **false flags**, and **reproduction completeness**. Report each case and uncertainty; twenty tickets are an exploratory pilot, not a corpus-wide accuracy benchmark.

Decision rule: revise or drop the tool if it adds time without additional confirmed findings or more reproducible evidence. Do not adopt it simply because the demo found a real conflict.

30 RECORDS. FIVE DISAGREEMENTS.

A targeted check of ITP 2023 DOI suffixes 1-30 found 25 normalized-title agreements and five disagreements. All 60 singleton responses ultimately returned HTTP 200; five transient failures were retried once and logged.

25 AGREE

5 DIFFER

DOI SUFFIXES WITH TITLE DIFFERENCES: 13 / 17 / 18 / 19 / 26

Selection followed discovery of the problem in this volume. It is a contiguous subset of 38 numbered contributions, not a random sample. **These counts are not an OpenAlex-wide error rate or a precision/recall evaluation.** They identify five review candidates, not five adjudicated merge failures.

Comparison used Unicode NFKC, case folding, and punctuation/whitespace normalization. Eleven creator-name lists also differed, but name formats can explain that; they are not counted as author errors. The cohort archive contains 70 raw receipts, including discovery and control checks.

RUN THE LIVE AUDIT / PYTHON STANDARD LIBRARY

```
python3 -m sourcecheck audit W4385245566 \  
  --output /tmp/sourcecheck-report.json \  
  --evidence-dir /tmp/sourcecheck-evidence
```

REPLAY THE ARCHIVED EVIDENCE / NO NETWORK

```
python3 scripts/verify_evidence.py  
python3 docs/verify_collision_evidence.py
```

The case verifier checks all eight receipt hashes and byte lengths, the publisher metadata, the five DOI locations, nine matching author ORCIDs, and the typed version/citation relations. The full verifier checks the archived cases and cohort calculations.

A later live response may differ because the public record has been corrected. Preserve the new receipt and report the change; do not force the live result to match this captured case.

COHORT + DISCOVERY LOG

PROOF YOU CAN INSPECT

Eight original response bodies independently captured on 17 September 2026, 17:22:15-17:22:17 UTC. All returned HTTP 200. Click a filename to inspect its live source. The repository manifest retains exact timestamps, URLs, status, byte counts, and hashes.

openalex-work.json	28,800 bytes
5529440446752467ff2c6b6f41e0f2098d8cf67aab1cd9673e1d6d35f146b69e	
datacite-mizar.json	31,620 bytes
dd3bd127b81660f435ce82674cbb416bcbf886f55dfb073380bf5f89bf0658cc	
datacite-tutoring.json	22,514 bytes
3e5299e95990dabd7dd13c57545a3dde9b2a9c8f49a29084662b2bab48ecc1fe	
datacite-zenodo-concept.json	12,251 bytes
5451b1fd0d7f92a54538a796043a53c5961e8bc2eafca244dfb78fc372c7a0e3	
datacite-zenodo-v1.json	12,190 bytes
f64eda439bb712006cc06e49d142eb01a04faa7972dc758542e14ba245b7868e	
datacite-zenodo-v2.json	12,242 bytes
add0cedee675ad00f1fd8a5135912788f3294fb57a61beebffc4cec7969a323	
publisher-mizar.html	176,794 bytes
65f9b50e26d765df5a540c073b77ff8d1075538a743903047835a7ab7d364a2e	
publisher-tutoring.html	108,186 bytes
f9df38c84ab9fe791d5c039e5c4164647fe14c47d61ccffee6b55e8b78dbff01	

WHAT THIS EVIDENCE DOES - AND DOES NOT - ESTABLISH

The captured work contains conflicting bibliographic identities. Registry and publisher agree on their DOI-specific claims, but may share the same metadata feed. Hashes verify captured bytes, not source truth. The exact internal cause, affected downstream fields, and appropriate upstream split require maintainer investigation.

No OpenAlex data was edited and no ticket was submitted. This is an independent application artifact, built with AI assistance. Public metadata, local Python, and free Vercel Hobby hosting are sufficient; no paid model or service is required.

[FULL TECHNICAL DOCUMENTATION](#)